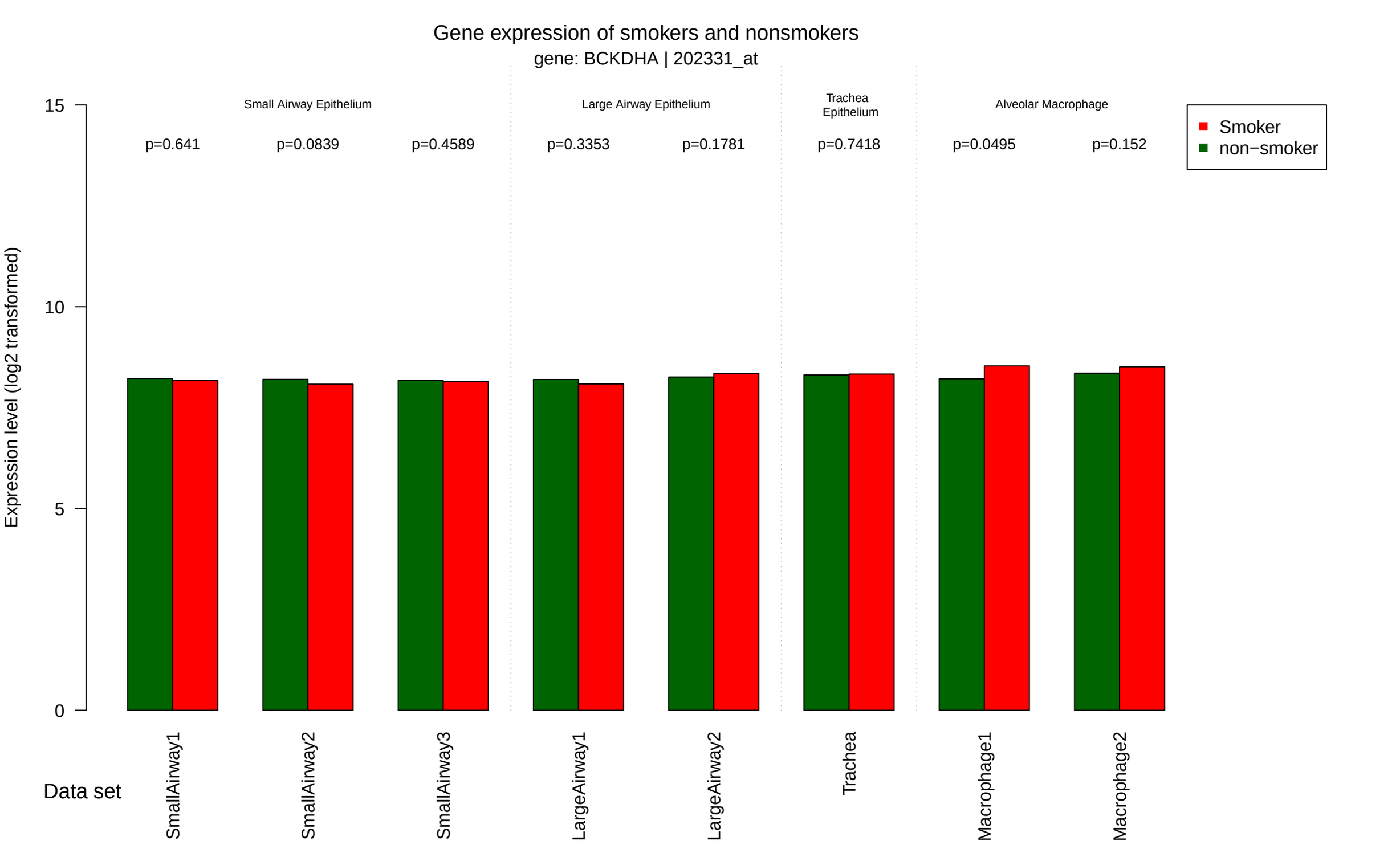
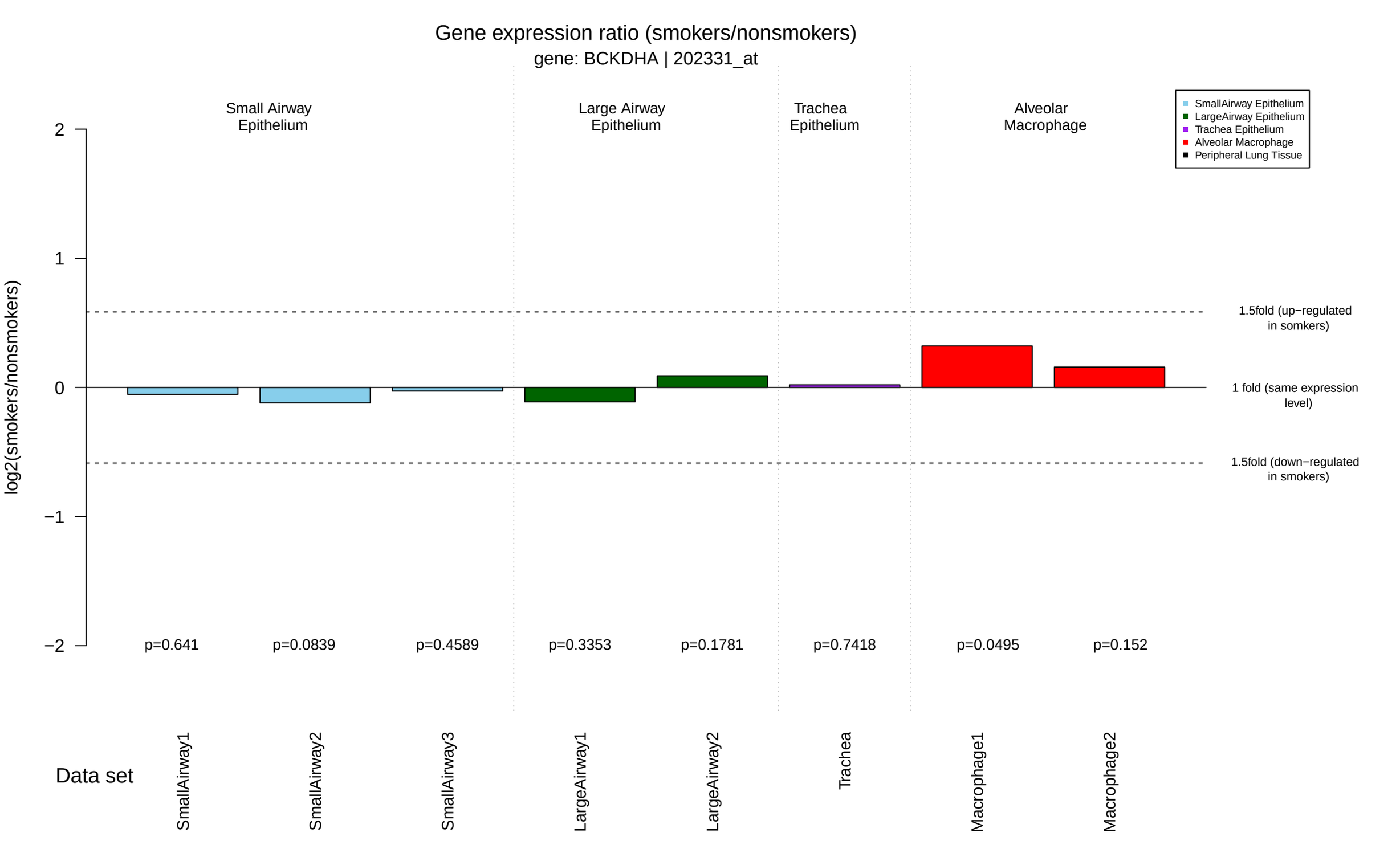
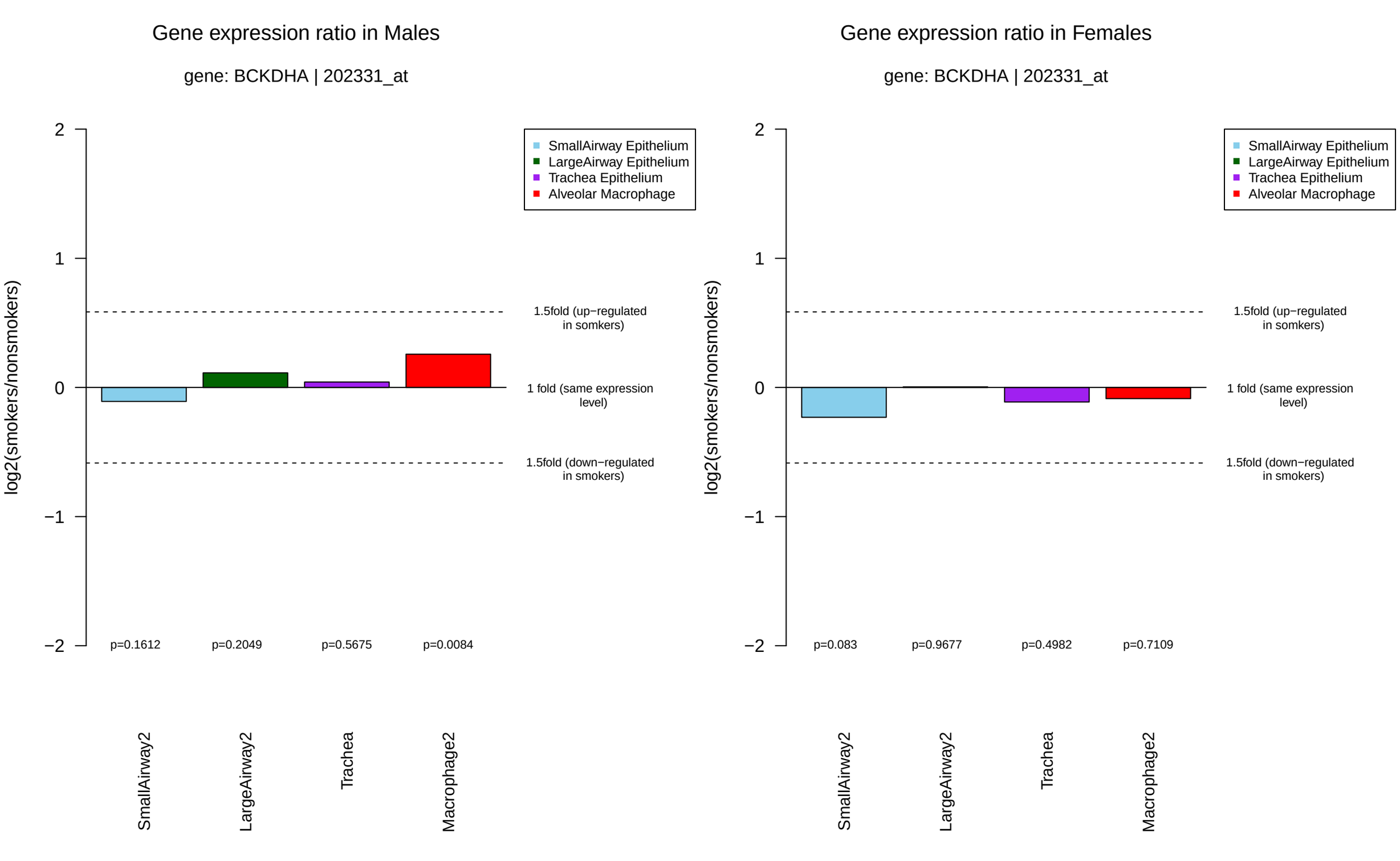
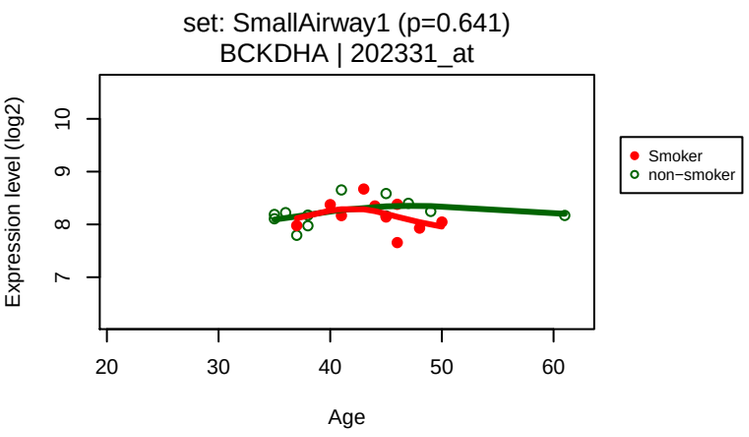
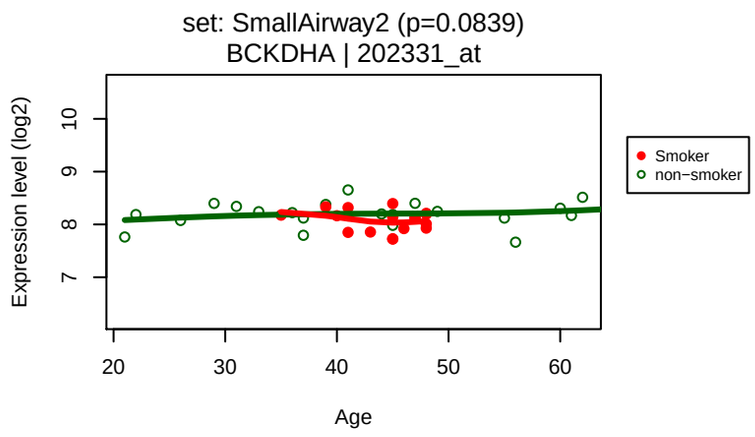
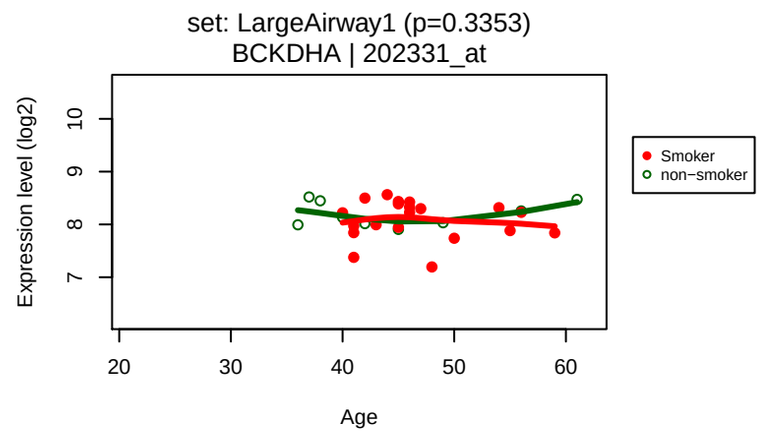

Gene expression of smokers and nonsmokers
gene: BCKDHA | 202331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.641
p=0.0839
p=0.4589
p=0.3353
p=0.1781
p=0.7418
p=0.0495
p=0.152
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

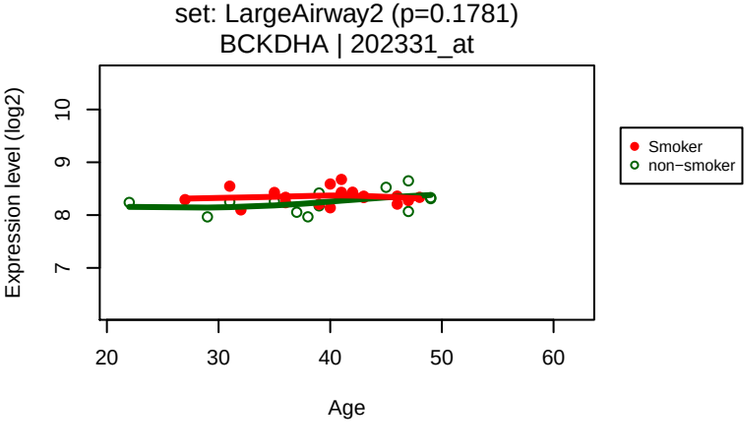

Gene expression ratio (smokers/nonsmokers)
gene: BCKDHA | 202331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.641
p=0.0839
p=0.4589
p=0.3353
p=0.1781
p=0.7418
p=0.0495
p=0.152
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

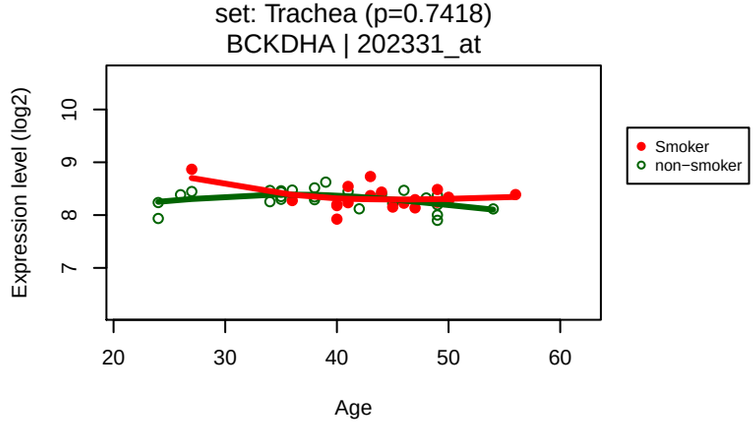

Gene expression ratio in Males
gene: BCKDHA | 202331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1612
p=0.2049
p=0.5675
p=0.0084
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCKDHA | 202331_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.083
p=0.9677
p=0.4982
p=0.7109
SmallAirway2
LargeAirway2
Trachea
Macrophage2

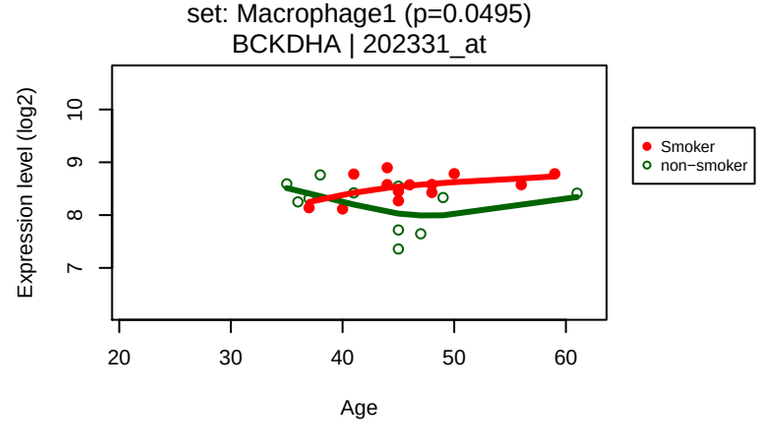

set: SmallAirway1 (p=0.641)
BCKDHA | 202331_at
Expression level (log2)
Age
Smoker
non−smoker

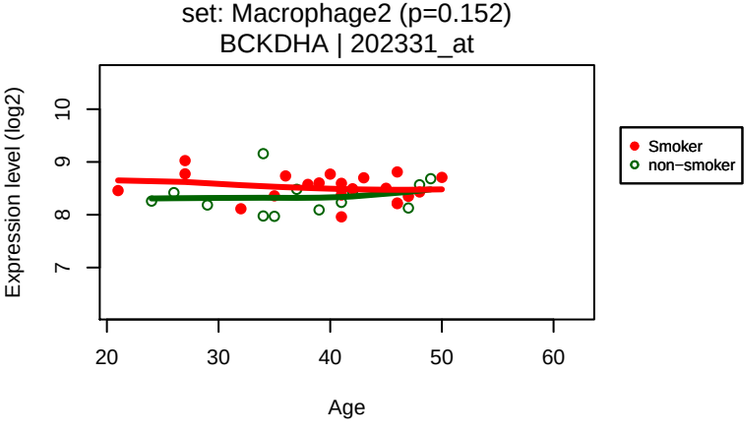

set: SmallAirway2 (p=0.0839)
BCKDHA | 202331_at
Expression level (log2)
Age
Smoker
non−smoker

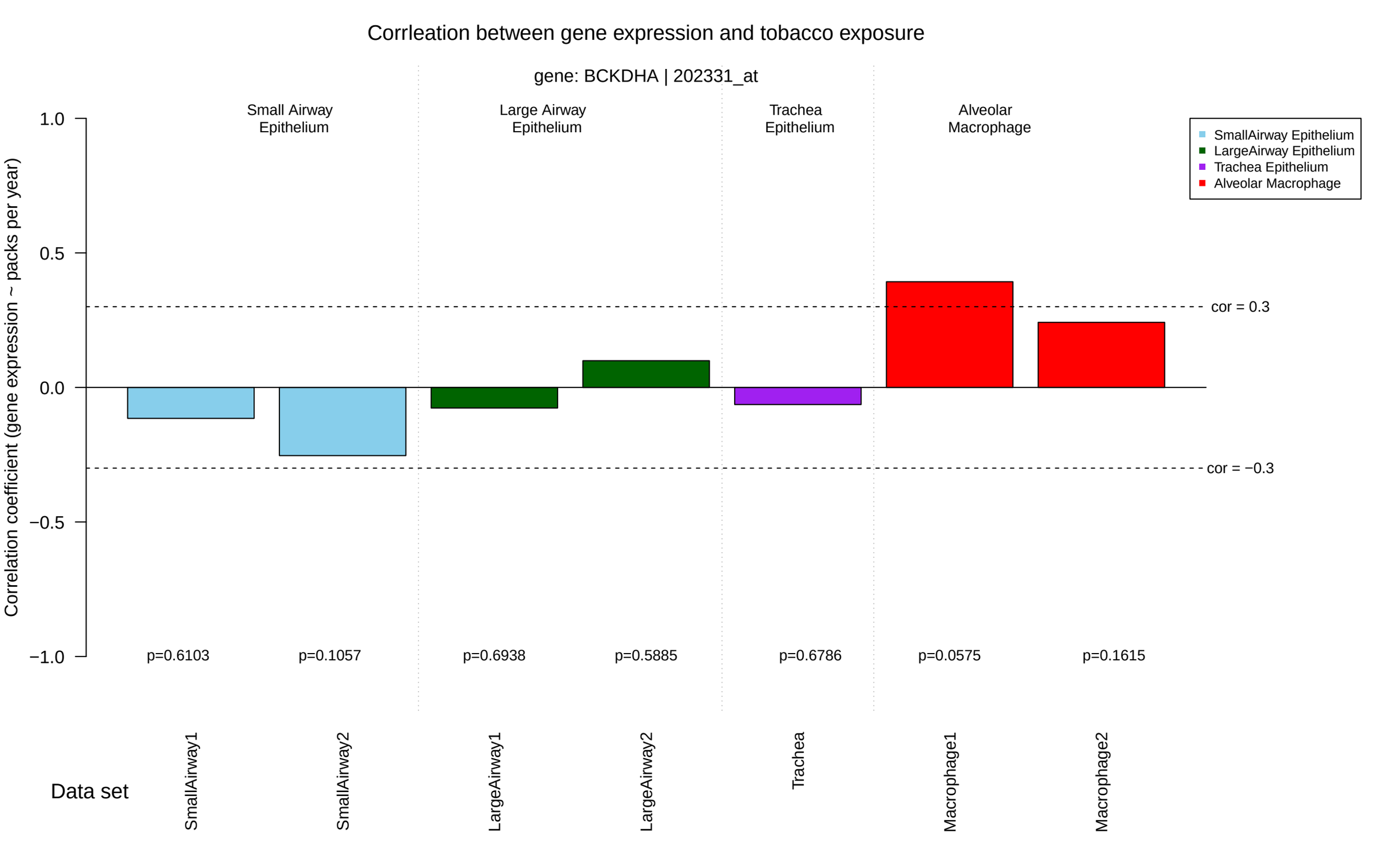

set: LargeAirway1 (p=0.3353)
BCKDHA | 202331_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1781)
BCKDHA | 202331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7418)
BCKDHA | 202331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0495)
BCKDHA | 202331_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.152)
BCKDHA | 202331_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCKDHA | 202331_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6103
p=0.1057
p=0.6938
p=0.5885
p=0.6786
p=0.0575
p=0.1615
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2